La Palma ML-FCUC — release verification report

Edition: 1.0.0-deployment, 14 October 2026. **Manuscript:** SEGAN-D-26-02403.

Release decision and scope

The public ZIP supplies all source-side dependencies for the **frozen-coefficient primary La Palma ML-FCUC optimization route**. It is not a new native GAMS execution certificate or a complete reconstruction of the original learning/evidence archive. The earlier source-recovery package is superseded by this edition.

The default entry is `GAMS_LaPalma/START_LAPALMA.gms`: one 24-hour LR_C1 winter/day-1 problem, then optional WEEK or ALL scope. The native requirements are a working Windows GAMS 24.9/CPLEX installation and GDXXRW/Excel export support. No classifier fitting, original LP_Case.json or LP35_Protocol.json is needed to call that optimizer.

Executed checks

Check	Result
Uploaded original numerical/coefficient include hashes	PASS — 28/28 match applicable frozen records
All primary direct numerical include dependencies	PASS — 18/18 present
Documented GAMS components versus original non-comment content	PASS — 7/7, with 26 public include comparisons
Primary equation definitions versus original and historical core	PASS — 35/35, whitespace-normalized only
Frozen primary calibrated include, job roster and CPLEX options	PASS — original byte hashes retained
Actual-fleet outage Big-M recomputation	PASS — 132/132 equal at original export precision
Frozen calibrated-include serialization	PASS — exact byte equivalence
Controlled scaler/calibrator-to-affine test	PASS — maximum probability difference 8.83e-13
Controlled parent-group fold test	PASS — three folds without parent leakage
Artificial SODE closed-form/refinement/limit checks	PASS — closed-form maximum error 3.21e-9 Hz
All ten public notebooks: structure, syntax and imports	PASS — import checks recorded separately
Four lightweight notebook executions	PASS — separate fresh kernels, then a separate ZIP extraction
ZIP integrity, extraction structure and unchanged files	PASS — CRC, safe single-root paths and file checks
Six original scientific notebooks	PASS for 46 code-cell-text preservation; full drivers NOT RUN
Exact originals preserved outside the public ZIP	PASS — 44 source/input/configuration files in PRIVATE backup

The result-reader notebook completed its input setup and numeric-reader checks. It found **zero native workbooks**, so its actual schedule audit correctly reports NOT RUN. The SODE fixtures are

explicitly artificial; they are not the accepted La Palma governor dataset or replacement manuscript results.

Solver provenance

The supplied cplex.opt is exactly `threads 4` plus a newline, and matches the original option-file digest. It does not override time or gap settings. The uploaded working main requests a 0.005 search target, 1,200-second search allowance and 300-second fixed-commitment polish allowance. Its original pre-run manifest records 0.0125 and 600 seconds. Reversing only those two defaults and restoring the original LF line ending reproduces the manifest's pre-run main-source hash.

The working settings are therefore identifiable and preserved. The earlier manifest is not edited. The prior embedded review of all 336 jobs is retained as historical evidence, not newly verified against inaccessible native listings. A source macro supplied by the user can still override a conditional default; the public start file does not supply an override.

Tests not performed / scope exclusions

NOT RUN: GAMS 24.9/CPLEX compilation or optimization, native Excel/GDX output generation, original production SODE labels, original classifier training/calibration, fitted-model equivalence on original training/test rows, full original 336-workbook/listing re-audit, and the six production notebook drivers. The solver installation and required full-research archives are not available in the builder. No heavy scientific jobs were run to create this package.

NOT APPLICABLE to the primary route: historical analytical SP reproduction or a new locational/network model. The historical SP include pair and standalone Raja reference package are not redistributed pending third-party permission. Numerical inputs and coefficient rows of the primary optimizer are unchanged.

Reading the status labels

PASS means the named test actually completed and satisfied its stated criterion. FAIL means the named test was attempted and its criterion was not met. NOT RUN means execution was not attempted or could not be performed with available dependencies. NOT APPLICABLE means the operation is outside this release's stated primary scope. There are no outstanding failed checks in the executed matrix; NOT RUN is not converted into PASS.

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Detailed records and independent rerun

The machine-readable verification matrix, source/code-cell hashes, equation definitions, dependency graph, source snapshot comparison and per-notebook results accompany this report. Run `notebooks/00_Quick_Start.ipynb`, `01_Frozen_Cuts_and_Export.ipynb` and `02_SODE_Equation_Checks.ipynb` from a fresh extraction. These checks require no external model fitting or solver. Native result auditing is enabled only after real native workbooks exist. Archive integrity and fresh-extraction checks are recorded in the final package verification metadata.